



Módulo 2 - Semana 5
Optimal PV and BESS sizing
Integración de FNCER

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Abril de 2026



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Resumen

This paper presents a novel Mixed-Integer Non-Linear Programming (MINLP) optimization model for the simultaneous sizing of Photovoltaic (PV) systems and Battery Energy Storage Systems (BESS) in *behind-the-meter* (BTM) self-consumption applications, specifically targeting a near-total independence from external grid imports. Unlike traditional sizing heuristics, the proposed formulation maximizes the project's Net Present Value (NPV) from a prosumer perspective by fully integrating hourly battery scheduling. The model's performance is validated against the standard simplified method presented by [1, 2]: reducing required storage capacity from 38.0 kWh to 30.0 kWh. Furthermore, the optimized dispatch management achieves a Levelized Cost of Electricity (LCOE) of 90.4 \$/MWh, representing a 14 % reduction compared to the 103.3 \$/MWh. of the simplified version. These findings underscore the pivotal role of advanced dispatch modeling in enhancing the economic viability and energy autonomy of large-scale renewable behind-the-meter applications.



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1. Introduction

The global transition toward decentralized energy has accelerated the adoption of behind-the-meter (BTM) solutions. These systems, primarily centered on photovoltaic (PV) generation, have reached unprecedented levels of cost-competitiveness due to rapid technological advancements and the plummeting capital costs of Battery Energy Storage Systems (BESS). This economic shift has transformed PV-plus-storage from a peak-shaving tool into a viable alternative for 24-hour baseload power supply in industrial and commercial sectors.

The optimization of photovoltaic (PV) generation and battery energy storage systems (BESS) has emerged as a cornerstone in the design of behind-the-meter (BTM) energy architectures. Recent scholarship indicates a paradigm shift from traditional rule-based sizing heuristics toward integrated optimization formulations. These modern approaches simultaneously determine PV and BESS capacities while internalizing high-resolution load variability, electrochemical degradation, and complex operational constraints.

A significant benchmark in this domain is the Ember 24-hour solar study [1], which demonstrates that near-continuous solar electricity is technically feasible in high-irradiance regions. By sizing assets to align with hourly demand-supply patterns, the study frames the sizing problem around stringent energy-autonomy targets rather than aggregate annual energy balances. Building upon this foundation, contemporary research from 2024–2025 has focused on enhancing these models through probabilistic frameworks, advanced forecasting, and integrated dispatch strategies. For instance, the Ember’s study demonstrates that continuous solar generation is technically feasible in Las Vegas, Nevada, USA at a Levelized Cost of Electricity (LCOE) of approximately 104 US\$/MWh [1]. Their analysis suggests that a 1 MW datacenter can achieve 97% energy independence using a heuristic sizing of 6 MW of PV and 17 MWh of storage. While these benchmarks are favorable when compared to traditional industry indicators [3], they often rely on static sizing methods that may not fully capture the economic potential of the system.

This paper we present a Mixed-Integer Non-Linear Programming (MINLP) model designed to determine the optimal configuration of PV and BESS assets by maximizing the project’s Net Present Value (NPV). Unlike traditional simplified sizing approaches [2], our model simultaneously optimizes capital investment (CAPEX) and operational scheduling (OPEX) while adhering to strict regulatory and technical constraints. Specifically, the model respects non-export protocols and maintains energy import limits within a predefined threshold, reflecting the “self-consumption reality of many modern prosumers.

The proposal is tested for a 2 kW datacenter located in Rio Hacha, Colombia (La Guajira). The remainder of this paper is structured as follows: Section 2 details the mathematical



formulation of the proposed MINLP model. Section 3 describes the case study parameters based on the Spanish regulatory framework, with the corresponding results and comparative analysis presented in Section 4. Finally, Section 5 provides the concluding remarks and future research directions.

2. Methodology

The proposed methodology provides a mathematical framework to determine the optimal specifications for PV and BESS components. The objective is to maximize the project's Net Present Value (NPV) from a prosumer perspective, considering a specific capital recovery factor and the operational realities of the energy market.

2.1. Optimization Framework

The problem is formulated as a Mixed-Integer Non-Linear Programming (MINLP) model, designed for high-performance solvers such as *Gurobi*. The objective function, defined in Eq. 1, seeks to maximize the NPV by balancing investment costs against long-term operational expenditures.

$$\text{máx NPV} = - \left(\frac{\text{OPEX}}{\text{CRF}_e} + \text{CAPEX} \right) \quad (1)$$

Where the NPV metric accounts for capital investment (CAPEX) and operation and maintenance costs (OPEX). The maximization of this objective is subject to two primary categories of constraints:

1. Economic Constraints: Governing financial parameters, including capital recovery, cost escalation, and maintenance expenditures.
2. Technical Constraints: Ensuring physical and operational feasibility, including hourly power balance, PV generation limits, BESS state-of-charge (SoC) dynamics, and grid interconnection limits.

To enhance clarity, all decision variables (unknowns) to be determined by the optimizer are denoted in blue throughout this section.

2.2. Parameters and Dataset Definitions

The model utilizes a comprehensive dataset spanning technical, economic, and regulatory parameters. We consider a prosumer entity—defined as an agent capable of both consumption



and self-generation—operating under a regulatory framework that permits surplus energy management but restricts or regulates grid injections.

The financial and technical parameters utilized in the formulation are summarized in Table 1.

Tabla 1: Model Parameters and Definitions

Parameter	Description
CRF	Capital Recovery Factor: $\frac{i(i+1)^n}{(i+1)^n - 1}$, where i is the discount rate and n the project lifespan.
CRF_e	Modified CRF incorporating the annual escalation factor e and modified discount rate i_r : $\frac{(1+e)i_r}{(i_r+1)}$.
T	Temporal resolution for power dispatch (typically 8,760 hours).
C_{pv}, C_{bess}	Overnight investment costs for PV (per kW) and BESS capacity (per kWh).
C_{inv}	Load-following inverter overnight cost used in PV and BESS applications in US\$/kW.
$O\&M_{pv}$	PV system operation and maintenance cost in US\$/kW.
$O\&M_{bess}$	BESS operation and maintenance cost in US\$/kWh.
BoP	Balance of plant, encompassing all components, infrastructure, and services excluding solar modules.
S_c	Project soft costs (permitting, design, etc.), typically 10–20 % of CAPEX.
$P_L^{inst}, P_{L,t}^u$	Installed load capacity (kW) and hourly load utilization factor (0–1).
$P_{pv,t}^u$	Average hourly PV system output (DC power injection) in per unit. Hourly solar generation data.
η_c, η_d	Efficiencies for BESS charging/discharging cycles.
P_F^{max}	Maximum power capacity at the frontier/interconnection point (kW).
β_{min}, β_{max}	BESS operational limits (SoC window); e.g., 0.05 to 0.95 for 90 % DoD.
C_R	BESS inverter C-rate limits (1/h), defining the minimum/maximum charge durations.

Proposed optimization model

The optimization problem is stated as finding the system OPEX (optimal energy and power dispatch: $\mathbb{P}_s, \mathbb{P}_b, \mathbb{P}_c, \mathbb{P}_d, \mathbb{P}_{pv}^{max}, \mathbb{P}_{pv}, W_s, W_b, W_{pv}, W_{pv}^{curtail}, SOC, SOC_0, \mathbf{w}, \mathbf{u}$) and CAPEX (equipment specification: $P_{pv}^{inst}, C_{bess}, P_{bess}^{inverter}, P_{pv}^{inverter}$) such that the net present value of the project is maximized.

$$\max_{\substack{P_{pv}^{inst}, C_{bess}, P_{bess}^{inverter}, P_{pv}^{inverter}, \mathbb{P}_b \\ \mathbb{P}_s, \mathbb{P}_c, \mathbb{P}_d, \mathbb{P}_{pv}, \mathbb{P}_{pv}^{max}, SOC, SOC_0 \\ W_s, W_b, W_{pv}, W_{pv}^{curtail} \\ \mathbf{w}, \mathbf{u}}} NPV = -\frac{OPEX}{CRF_e} - CAPEX \quad (2)$$

subject to:



$$OPEX = O\&M_{pv}P_{pv}^{inst} + O\&M_{bess}C_{bess} \quad (3)$$

$$CAPEX = (C_{pv}P_{pv}^{inst} + C_{bess}C_{bess} + C_{inv}(P_{bess}^{inverter} + P_{pv}^{inverter}))S_C + BoP \quad (4)$$

$$P_{d,t} + P_{b,t} + P_{pv,t} = P_{c,t} + P_{s,t} + P_{L,t}^u P_L^{inst}, \forall t = 1, \dots, T \quad (5)$$

$$P_{pv,t}^{max} = P_{pv,t}^u P_{pv}^{inst}, \forall t = 1, \dots, T \quad (6)$$

$$P_{pv}^{inverter} = P_F^{max} + P_L^{inst} + P_{bess}^{inverter} \quad (7)$$

$$P_{pv,t} \leq P_{pv,t}^{max}, \forall t = 1, \dots, T \quad (8)$$

$$P_{c,t} \leq P_{bess}^{inverter} w_t, \forall t = 1, \dots, T \text{ [Mixed-Integer (Binary) Non Linear constraint]} \quad (9)$$

$$P_{d,t} \leq P_{bess}^{inverter} (1 - w_t), \forall t = 1, \dots, T \text{ [Mixed-Integer (Binary) Non Linear constraint]} \quad (10)$$

$$P_{b,t} \leq P_F^{max} u_t, \forall t = 1, \dots, T \quad (11)$$

$$P_{s,t} \leq P_F^{max} (1 - u_t), \forall t = 1, \dots, T \quad (12)$$

$$SOC_t = SOC_{t-1} + P_{c,t}\eta_c - \frac{P_{d,t}}{\eta_d}, \forall t > 1, \quad SOC_1 = SOC_0 + P_{c,1}\eta_c - \frac{P_{d,1}}{\eta_d} \quad (13)$$

$$\beta_{min}C_{bess} \leq SOC_t \leq \beta_{max}C_{bess}, \forall t = 0, 1, \dots, T \quad (14)$$

$$C_R^{min}C_{bess} \leq P_{bess}^{inverter} \leq C_R^{max}C_{bess} \quad (15)$$

$$W_s = \sum_{t=1}^T P_{s,t}, \quad W_b = \sum_{t=1}^T P_{b,t}, \quad W_{pv} = \sum_{t=1}^T P_{pv}^t, \quad W_{pv}^{curtail} = \sum_{t=1}^T P_{pv,t}^{max} - W_{pv} \quad (16)$$

$$W_s = 0 \quad (17)$$

$$W_s = \alpha \sum_{t=1}^T P_{L,t}^u P_L^{inst} \quad (18)$$

The optimization is formulated as a continuous 8,760-hour dispatch model to determine the project's Net Present Value (NPV) over a lifespan of n years at a specified discount rate i_r . Results are evaluated using discounted energy and costs based on a standard annual timeframe, incorporating the PVGIS-ERA time series data for the period 2005–2023.

The resulting Mixed-Integer Non-Linear Programming (MINLP) model comprises $8 + 9T$ unknown variables, $10 + 3T$ equality constraints, and $10 + 11T$ inequality constraints. While the degrees of freedom are established by the fact that variables outnumber equality constraints, the extensive set of inequality constraints significantly delimits the feasible region. The decision variables are categorized into four primary groups: PV and BESS equipment specifications, economic parameters, and power/energy dispatch outcomes.



Equipment Specifications

The primary sizing decision variables for the optimization problem are defined as:

P_{pv}^{inst} : Installed capacity of the PV system [kW].

C_{bess} : Total battery energy storage capacity [kWh].

$P_{bess}^{inverter}$: Power rating of the BESS inverter [kW].

$P_{pv}^{inverter}$: is the PV inverter capacity [kW].

As an illustrative example, the required number of modules (N_{pv}) for a system utilizing 30 W crystalline silicon technology—under a standard maximum irradiance of $1,000 \text{ W/m}^2$ —is determined by:

$$N_{pv} = \frac{P_{pv}^{inst} \cdot 1000}{1000 \cdot A \cdot \eta} \quad (19)$$

where $A = 0,23 \text{ m}^2$ represents the area of a single PV module and $\eta = 14\%$ denotes the module efficiency.

Economic Variables

OPEX: Annual operational expenditure for power and energy imported from the market following the PV & BESS installation [US\$/year].

CAPEX: Capital expenditure associated with the initial investment in PV and BESS equipment [US\$].

Optimal Power and Energy Dispatch Variables

$\mathbb{P}_{pv}$ = $[P_{pv,1}, \dots, P_{pv,T}]$: Vector of AC power injected by the PV inverter at hour t [kW].

$\mathbb{P}_{pv}^{max}$ = $[P_{pv,1}^{max}, \dots, P_{pv,T}^{max}]$: Vector of potential DC power generated by the PV system at hour t [kW].

$\mathbb{P}_s$ = $[P_{s,1}, \dots, P_{s,T}]$: Vector of power exported to the external grid at hour t [kW].

$\mathbb{P}_b$ = $[P_{b,1}, \dots, P_{b,T}]$: Vector of power imported from the external grid at hour t [kW].

$\mathbf{u}$ = $[u_1, \dots, u_T]$: Binary vector indicating grid status (1 for importing, 0 for exporting).



$\mathbb{P}_c = [P_{c,1}, \dots, P_{c,T}]$: Vector of BESS charging power at hour t [kW].

$\mathbb{P}_d = [P_{d,1}, \dots, P_{d,T}]$: Vector of BESS discharging power at hour t [kW].

$\mathbf{w} = [w_1, \dots, w_T]$: Binary vector indicating BESS status (1 for charging, 0 for discharging).

$\text{SOC} = [\text{SOC}_1, \dots, \text{SOC}_T]$: Vector of the state of charge at hour t [kWh].

SOC_0 : Initial state of charge [kWh].

W_s, W_b, W_{pv} : Annual energy exported, imported, and generated by the PV system, respectively [kWh/year].

W_{pv}^{max} : Total annual DC PV energy generation, where $W_{pv}^{max} = \sum_{t=1}^T P_{pv,t}^{max}$.

$W_{pv}^{curtail}$: Annual energy curtailed by the PV system [kWh/year].

The optimization model is governed by the following constraints:

- Financial Metrics: Eq. 3 and Eq. 4 define the project's capital and operational expenditures, respectively.
- Power Balance: Eq. 5 enforces the real power balance at the point of interconnection (POI).
- PV inverter specification: Eq. 7 specifies the size of the PV inverter in kW. The PV inverter capacity should be enough larger to cover loads, the BESS system and the point of interconnection. Inverter Loading Ratio (ILR), also known by the DC-to-AC Ratio ($ILR = P_{pv}^{inst} / P_{pv}^{inverter}$) represents the proportion between PV installed capacity at standard test conditions and PV inverter output.
- Generation Limits: Eq. 6 ensures PV inverter output is limited by rated capacity, efficiency, and real-time solar resource availability.
- Storage Constraints: Eqs. 9–10 restrict BESS charging and discharging rates to the inverter's capacity. These are *non-linear* due to the product of binary variables and sizing capacities.
- Grid Interface: Eq. 11 and Eq. 12 limit power imports and exports based on the physical feeder/transformer capacity at the boundary.



- Battery Dynamics: Eq. 13 models the energy balance and state of charge transition, incorporating distinct efficiencies for charging and discharging. It also enforces a cyclic boundary condition ($SOC_T = SOC_0$).
- Operational Ranges: Eq. 14 maintains the SOC within safety limits, while Eq. 15 ensures BESS power and capacity adhere to specified C-rate limits.
- Energy Totals: Eq. 16 aggregates annual energy metrics and quantifies curtailment.
- Regulatory Compliance: Eq. 17 prohibits the export of PV-generated energy to the grid, and Eq. 18 restricts grid imports to a specific sharing coefficient α (e.g., 5.5 %).

Economic Assessment Metrics

The Net Present Value (NPV) serves as the primary metric to evaluate the economic viability of the project from the prosumer's perspective, accounting for both the initial Capital Expenditure (CAPEX) and the multi-year Operational Expenditure (OPEX). To provide a comprehensive financial analysis, this is supplemented by the Internal Rate of Return (IRR) and the Levelized Cost of Electricity (LCOE).

The LCOE represents the average cost per unit of energy generated over the project's lifetime, defined as:

$$LCOE = \frac{CAPEX + \frac{OPEX_{O\&M}}{CRF_e}}{\frac{W_L}{CRF}} \quad (20)$$

In this context, $OPEX_{O\&M}$ refers specifically to fixed operation and maintenance costs, calculated as:

$$OPEX_{O\&M} = O\&M_{pv} \cdot P_{pv}^{inst} + O\&M_{bess} \cdot C_{bess} \quad (21)$$

From the utility and investor perspective, the LCOE is fundamentally a positive value that indicates the break-even energy price required to achieve a null NPV. This calculation considers a project horizon of n years, a discount rate i , and an energy escalation rate e .

3. Case Study: 2 kW Data Center in Rio Hacha Colombia

Simplified analysis for a datacenter in Rio Hacha, Colombia, with a peak load (P_L^{inst}) of 2 kW (unitary load factor) reported an installed PV capacity (P_{pv}^{inst}) of 10.1 kW and a



battery capacity (C_{bess}) of 38.0 kWh, maintaining an annual grid dependency of only 5.5 % according to PVsyst simulation [2].

In this section, we describe the dataset required to apply the proposed method.

3.1. Load Characteristics

The study focuses on a 2 kW data center located in Rio Hacha (11,532°N, 72,932°W), Colombia (La Guajira). The facility is connected to a 220 V low-voltage AC busbar. The annual energy consumption (W_L) is 17,520 kWh, reflecting a constant load factor ($P_{L,t}^u = 1$) for all $t = 1, \dots, 8760$. The existing interconnection infrastructure has a maximum capacity (P_F^{max}) of 2 kW, which is deemed sufficient for the proposed project without requiring further upgrades.

3.2. Economic and Technical Parameters

We adopt the economic assumptions specified in the Ember benchmark [1]. The project lifespan (n) is 20 years with an annual discount rate (i) of 7.7 %. The Capital Recovery Factor (CRF) is determined as:

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} = 0,099589$$

Accounting for an annual energy price escalation factor (e) of 2.5 %, the modified capital recovery factor (CRF_e) is 0.082760.

The PV investment cost (C_{pv}) is set at 436 US\$/kW, based on the following breakdown:

- Modules: 110 US\$/kW, reflecting the late 2024 price range for TOPCon modules (90–120 US\$/kW).
- Balance of System (BoS): 143 US\$/kW, covering cabling, mounting, and monitoring [4].
- Installation: 135 US\$/kW for mechanical and electrical assembly [4].

BESS investment costs are defined as $C_{bess} = 185$ US\$/kWh for energy capacity and $C_{inv} = 48$ US\$/kW for the bi-directional inverter. These figures align with 2024 global estimates, though some reports suggest utility-scale costs are trending as low as 72 US\$/kWh [5]. Soft costs (S_c), including design and permitting, are accounted for via a 20 % multiplier ($S_c = 1,2$) applied to the total CAPEX.



The per-unit PV generation profile ($P_{pv,t}^u$) is derived from the PVGIS-ERA5 dataset (2005–2023), see Fig. 1. The simulation assumes optimized tilt and azimuth angles for the Rio Hacha region, incorporating a standard system loss factor of 16.5 % (include PV inverter losses).

Latitude (decimal degrees): 8.787

Longitude (decimal degrees): -75.858

Elevation (m): 16

Radiation database: PVGIS-ERA5

Slope: 13 deg. (optimum)

Azimuth: -21 deg. (optimum)

System losses (%): 21.3 (PV include inverter losses)

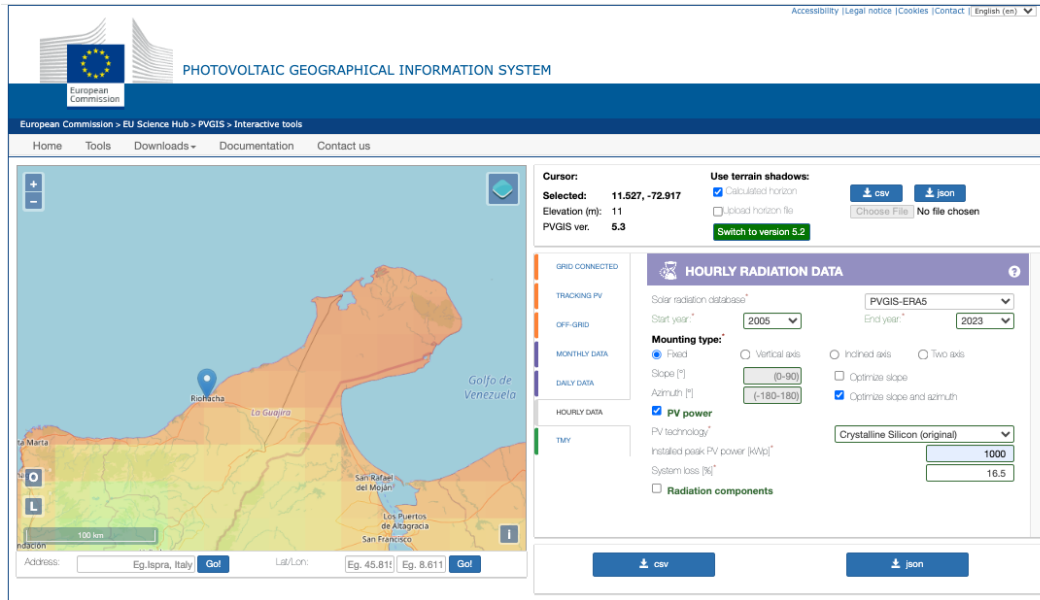


Figura 1: Portal de PVGIS https://re.jrc.ec.europa.eu/pvg_tools/en/

Regarding BESS operations, the Depth of Discharge (DoD) is set at 90 %, constraining the State of Charge (SOC) between $\beta_{min} = 0,05$ and $\beta_{max} = 0,95$. The BESS C-rate is bounded between 0.1 and 2.0.

Component efficiencies are modeled as $\eta_c = 95 \%$ and $\eta_d = 95 \%$.



4. Results and Discussion

The proposed MINLP optimization model, defined by Eqs. 2–17, was implemented in Python using the Gurobi Optimizer (v12.0.1) on an Intel Core i7-6700HQ (2.60GHz) workstation. The source code and datasets are available via GitHub¹.

The computational complexity of the 8,760-hour dispatch model is characterized by 78,872 decision variables (including 17,520 binary variables) and approximately 87,600 constraints. Despite the large-scale nature of the MINLP problem, an optimal solution with a relative gap of 0.00 % was achieved in 338.72 seconds.

4.1. System Sizing and Financial Performance

The optimization results for $\alpha = 5.5\%$. This value is taken from the PVsyst simulation for 10.1 kW and 38 kWh specified in [2] whose details are provided in the appendix. In this case, the optimized procedure indicates that a balanced sizing of 10.3 kWdc for the PV system and 30.0 kWh for the BESS is optimal for the 2 kW data center load. The model selects a PV inverter of 8.8 kW (ILR=1.16). This implies the installation of 343 PV-30W modules (poly).

A BESS inverter capacity of 4.5 kW is required, yielding a C-rate of 0.16 1/h, 6.21 hours. The battery capacity (30.0 kWh) is considerably smaller than the specified by the PVSyst platform (38.0 kWh).

Financially, the project requires a total capital expenditure (CAPEX) of 12,220 US\$ and an annual operational expenditure (OPEX) of 305 US\$. The resulting Levelized Cost of Electricity (LCOE) is 90.4 US\$/MWh. As summarized in Table 2, the proposed model significantly outperforms the benchmark sizing model developed in [2].

¹https://github.com/pmdeoliveiradejesus/MISE_P3

Tabla 2: Comparative analysis of PV & BESS sizing: Simplified benchmark vs. Proposed model.

Variable	Unit	Simplified Model [2]	Proposed Model
BESS capacity (C_{bess})	kWh	38.0	30.0
BESS inverter (P_{bess}^{inv})	kW	4.5	4.8
PV system (P_{pv}^{inst})	kW	10.1	10.3
PV inverter (P_{pv}^{inv})	kW	8.5	8.8
Net Present Value (NPV)	US\$	-18,123	-15,907
CAPEX	US\$	13,888	12,220
OPEX	US\$/yr	350	305
LCOE	US\$/MWh	103.0	90.4

The optimized approach achieves a 12.0 % reduction in CAPEX and a 13 % reduction in OPEX. The NPV increased 14 %. While the simplified sizing model [2] relies on a smaller PV array (10.1 kW) but a larger battery (38 kWh), the proposed model utilizes a higher-capacity BESS inverter. This suggests that the ability to rapidly charge and discharge the battery is more economically advantageous than simply increasing the storage volume, allowing for better management of high-irradiance peaks and load fluctuations.

4.2. Energy Dispatch

The annual energy balance is illustrated in Figure 2. The model ensures that the condition $W_d - W_c + W_b + W_{pv} - W_L = 0$ is strictly met. With a total annual load of 17.54 GWh, the system successfully maintains a grid dependency of only 5.5 % ($W_b = 963.60$ kWh/year). Available collector energy at STC is 22,819 GWh. System losses are 16.5 % (including 92 % inverter efficiency).

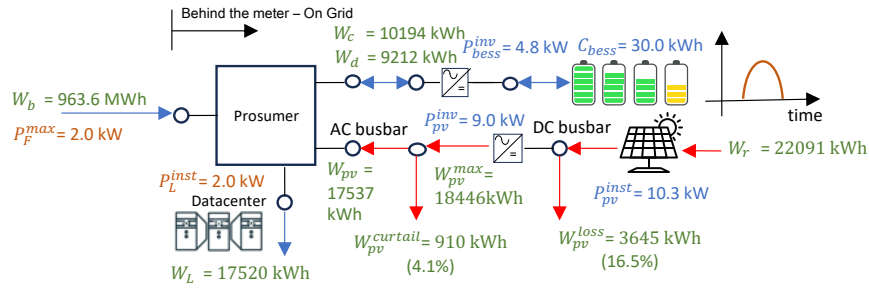


Figura 2: Optimized annual energy balance showcasing generation, storage, and grid import ratios.

It is worth to note that curtailment achieves a high value of 6.6 % with respect to collector energy.

The daily contribution to the 48 kWh load is shown in Figure 3. Notice that during the summer months, the system achieves 100 % energy independence, requiring zero imports from the grid. Conversely, grid imports are concentrated in the winter months when solar resource availability is at its minimum.

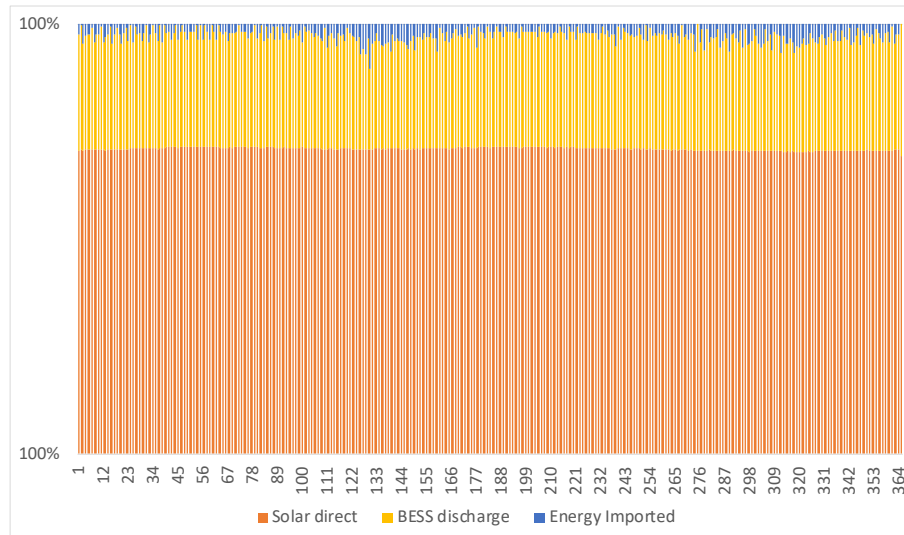


Figura 3: Daily demand components by source: Direct Solar, BESS Discharge, and Grid Import.

Figure 4 provides a granular view of the BESS behavior during typical weeks in January and July.

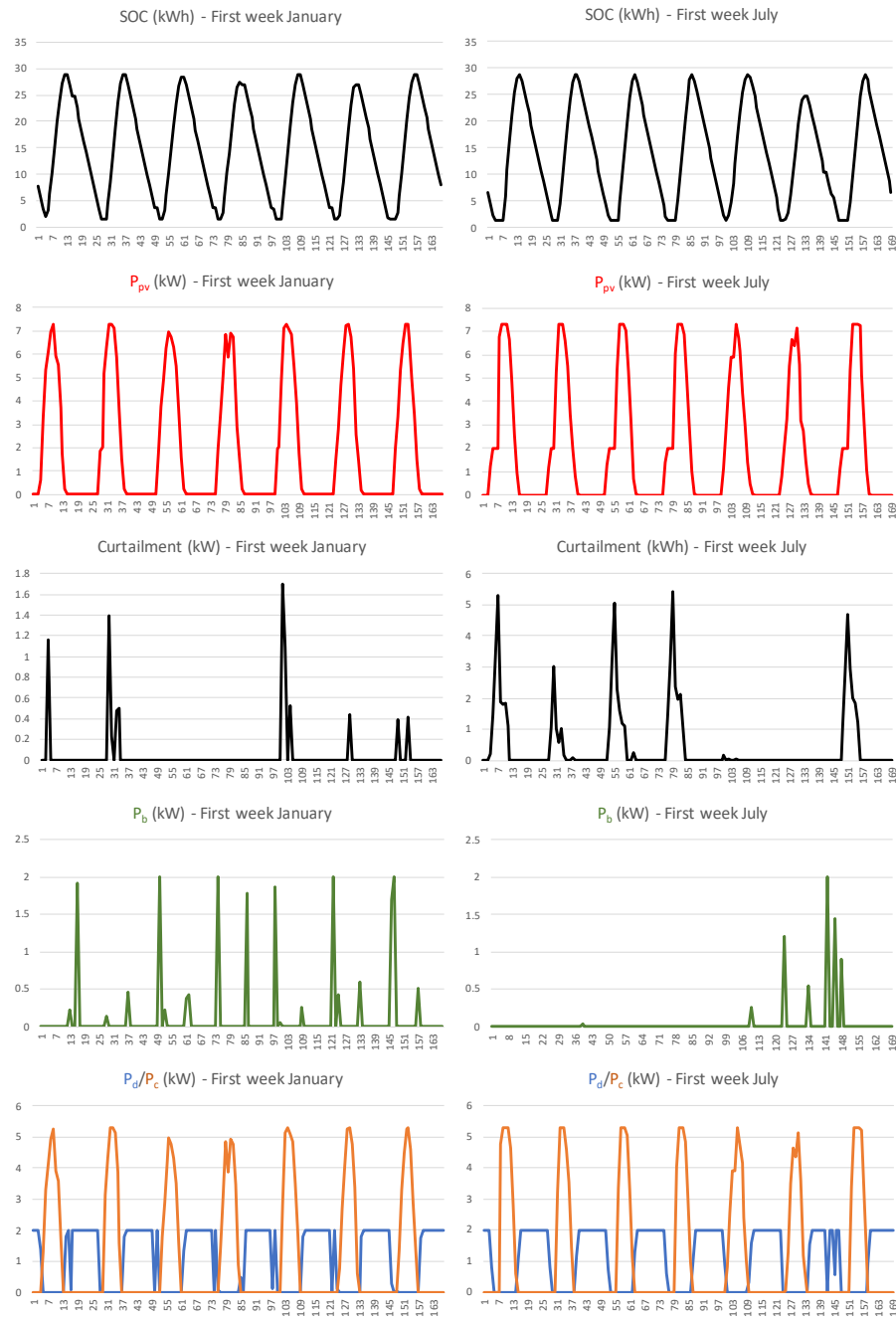


Figura 4: BESS State of Charge (SOC) and power dispatch behavior

The results show that sizing PV and BESS systems through integrated resource assessment and optimal dispatch management makes prosumer-led self-consumption highly competi-



ve. The proposed model achieves a LCOE of 90.4 US\$/MWh—a 14 % improvement over the benchmark (103.30 US\$/MWh) while ensuring a constant 24/7 power supply with 94.5 % reliability from renewable sources. This underscores the importance of inverter power-to-energy ratios and intelligent scheduling in the economic design of behind the meter applications.

A fundamental limitation of the model is that the cost of imported energy W_b is not considered. The model should incorporate earnings and costs of energy trading with the external grid according to Colombian regulation. In this sense, the resulting economic results are slightly overestimated.

5. Conclusion

This paper demonstrates that the optimal sizing of behind-the-meter (BTM) energy systems—specifically those integrating Photovoltaic (PV) and Battery Energy Storage Systems (BESS) for self-consumption applications aiming for near-total independence from grid imports—is linked to operational asset management. By incorporating a high-resolution, 8,760-hour MINLP dispatch framework that accounts for BESS scheduling and strategic PV curtailment, we move beyond static sizing approximations toward a more cost-effective and autonomous energy architecture.

The validity of the proposed model was benchmarked against the simplified sizing methodology [2] using a 2 kW data center case study in Rio Hacha, La Guajira, Colombia. The results reveal that an optimized dispatch-aware approach enables a more lean system design:

- A slight increase in PV capacity from 10.1 kW to 10.3 kW.
- A decrease in energy storage requirements from 38.0 kWh to 30.0 kWh.
- A significant improvement in the Levelized Cost of Electricity (LCOE), falling from 103.0 \$/MWh to 90.4 \$/MWh.

Ultimately, this study proves that while the BTM transition is often viewed through the lens of capital cost reduction, the true economic potential is unlocked by maximizing the synergy between generation, storage, and demand through advanced mathematical optimization.



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A. Benchmark Analysis: Simplified sizing model

To establish a comparative baseline, the optimization model described in Section 2 was applied to the parameters reported in the simplified study [2]. In this scenario, the capacities for the PV system and BESS were fixed at $P_{pv}^{inst} = 10,1$ kW and $C_{bess} = 38$ kWh, respectively. This allows for an evaluation of how the proposed optimal dispatch management performs when restricted to the benchmark’s equipment sizing.

Figure 5 shows the Sankey diagram provided by PVsyst simulator for $P_{pv}^{inst} = 10,1$ kW and $C_{bess} = 38$ kWh at Rio Hacha location.

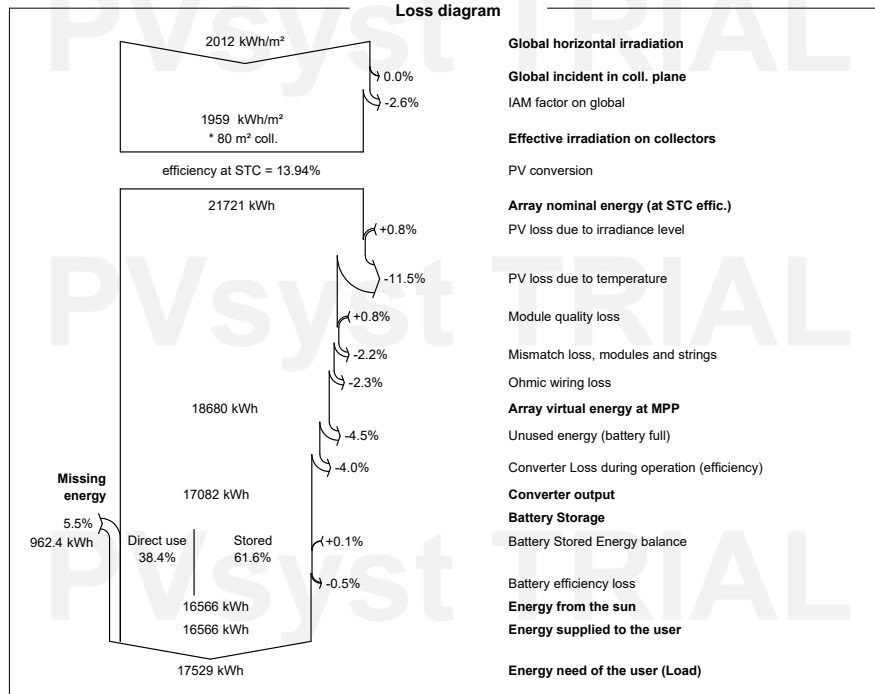


Figura 5: PVsyst simulator Sankey Diagram for $P_{pv}^{inst} = 10,1$ kW and $C_{bess} = 38$ kWh,

The solver achieved an optimal solution (gap $\leq 0.01\%$), with the resulting system performance metrics summarized below.

A.1. Benchmark Operational and Financial Results

Under the fixed capacity constraints, the model identifies the required BESS inverter capacity as 5.0 kW to manage the energy throughput efficiently. The PV inverter is set as 9.0 kW.

Operational Performance: ■ Annual Load (W_L): 17,520.00 kWh/year

- Grid Dependency (W_b): 963.6 kWh/year (5.5 % of total demand)
- BESS Throughput: 10,186 kWh charged vs. 9,225 kWh discharged.
- Energy Loss: The PV system generates 21712 kWh annually, but 3587 kWh (approx. 16.5 %) is curtailed due to the saturation of the 38 kWh battery and the limited inverter capacity.

Financial Outcomes: ■ CAPEX: 13,888 US\$.

- OPEX: 350 US\$/year.
- Net Present Value (NPV): −18,123 US\$.
- Levelized Cost of Electricity (LCOE): 103.3 US\$/MWh.

A.2. Energy Balance and Dispatch Visualizations

The annual energy balance for the benchmark case is illustrated in Figure 6. It highlights the significant role of curtailment in this oversized PV configuration.

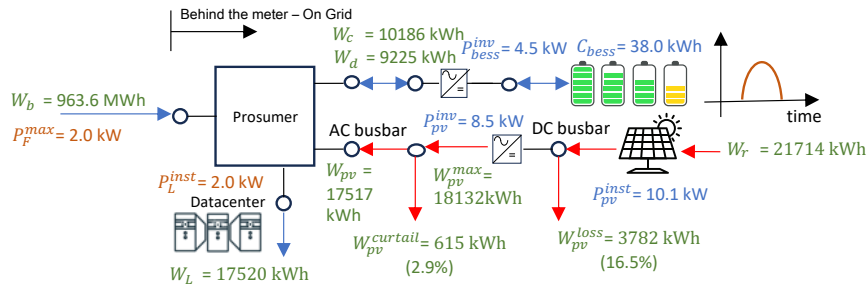


Figura 6: Annual energy balance of the project for the benchmark case (10,1 kW PV, 38 kWh BESS).

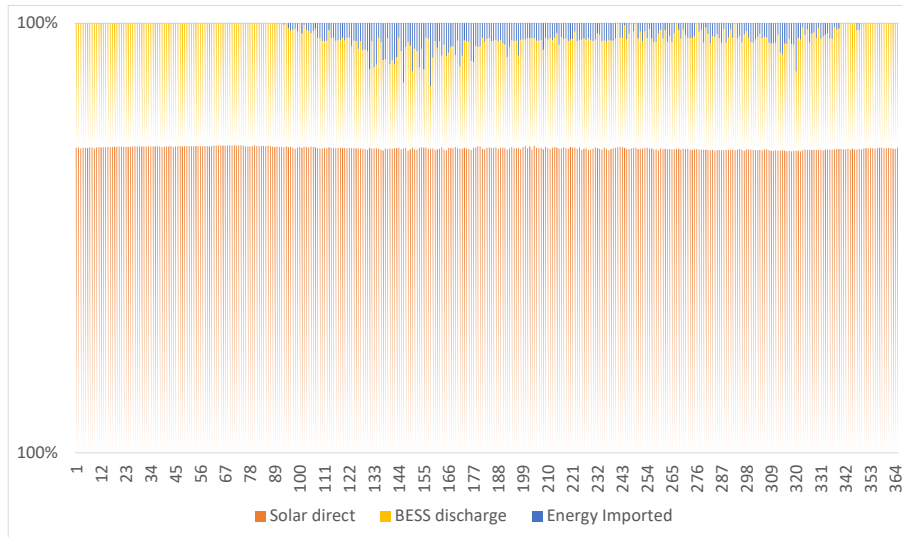


Figura 7: Annual energy dispatch for the fixed-capacity benchmark (Base Case).



Figure 7 depicts the annual dispatch profile. While the system successfully maintains the 5.5 % grid import target, the high level of curtailment suggests that the sizing is not economically optimal relative to the available solar resource in Rio Hacha.

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